Syllabus for CS 7150 Fall 2026

Class time and Location: Tuesdays 11:45 AM - 1:25 PM and Thursdays 2:50 PM - 4:30 PM, Behrakis Health Sciences Center 325 (BK 325)

The class meetings are in-person only.

The class is very participatory, with groups of students presenting and discussing papers on most days. On the first day of class, please fill out the Paper Presentation Panel Sign-up form to sign up.

Sign up for piazza here: https://piazza.com/northeastern/fall2026/cs715017500202710

Professor: David Bau davidbau@northeastern.edu

Professor Office Hours: right before class 2pm Thursdays at the lecture hall.

TAs: Grace Proebsting proebsting.g@northeastern.edu, Vishwajeet Hogale hogale.v@northeastern.edu

TA help hours:

To be posted.

Summary

In this course we will learn the principles and practice of deep learning methods and research.

We will cover the capabilities of deep networks, the main methods for training them effectively, and the common architectures and techniques for using a deep network to process and produce images and natural language text. In addition to getting experience using these methods, we will discuss seminal research papers, overview some of the main questions and debates that have emerged in deep learning research, and explore some current research topics. There will be a final project where you work with a small team to choose, investigate, and explain a deep learning research topic centered on a paper you choose.

How this course works

This course is about three things: **learning to learn**, **learning to teach**, and **teaching to learn**.

Learning to learn: picking up a tutorial or an unfamiliar paper, working out what it teaches, and learning to build what it describes. That is what the homeworks and exams are for. This semester there will be **five homework assignments**, each scored by an **in-class homework quiz**, and **one midterm exam**.

Learning to teach: explaining is a technical skill, and researchers spend much of their time on it. Papers, talks, diagrams, demonstrations and reviews are all ways of doing it, and you will practise all five. You will **read 19**

seminal papers and submit a discussion-sparking question for each one, and **present one of those papers** in class, creating an explanation and interactive visualization with a small team.

Teaching to learn: you will create a final project with a small team, and you will **present your own work** the same way you presented a seminal paper, providing context, technical explanations, and an interactive visualization explaining what you have learned.

The course is graded in a way to encourage you to do good work in the course, to keep learning even when things are hard, so that one bad day does not follow you around.

This course has been restructured for the LLM assistance era. LLM assistance is allowed and expected, but you must use it to enhance *your understanding* and your ability to explain the concepts of this course to your peers. Almost no points in the course are granted for work that an LLM can do (only 6/206 points are for completion rather than understanding); your goal is not just to produce some working code, but to learn the material well enough to work with the concepts on your own in quizzes and exams, to explain important concepts to your peers, and to gain insights and explain what you have learned when doing your group project.

Grading

There are **206 points** of course work in total.

Class participation and paper presentations	50
Homework quizzes (5 quizzes, 10 points each)	50
Midterm exam	50
Final project	50
Completion points (Quiz 0 and 5 homeworks, 1 point each)	6
Total	**206**

Tentative Class Schedule

(Note: we are sure to alter this schedule, likely to omit some topics or discuss others.)

DATE	*NOTE*	HANDOUT	DUE	TOPIC	ACTIVITY	READING
Thursday, September 10, 2026		**HW1, reading signups**	**Quiz 0**	**History: neurons, perceptrons, universality, backprop.**	**david**	**https://papers.baulab.info/**

Tuesday, September 15, 2026			Signups	Tensors, GPUs, Autograd, Optimizers, Modules, DataLoader	How-to-read-pytorch	Rumelhart 1986 (Backpropagation)
Thursday, September 17, 2026				Classification and backpropagation	david	Baydin 2018 (Autodiff survey)
Tuesday, September 22, 2026	***Last day to add/drop***	**HW2**	**HW1 + Quiz 1**	**Optimization and deep networks**	**david**	**Kingma 2015 (ADAM)**
Thursday, September 24, 2026				Initialization	**presentation**	Glorot 2010 (initialization)
Tuesday, September 29, 2026	*Prof. Bau away; Grace Proebsting hosts*			Equivariances and convolutions	**presentation**	LeCun 1989 (LeNet)
Thursday, October 1, 2026				Achieving Depth: residual nets, batchnorm	**presentation**	He 2016 (ResNet) *also see:* Ioffe 2015 (Batch normalization)
Tuesday, October 6, 2026		**HW3, project signup**	**HW2 + Quiz 2**	**Neural Language modeling**	**david**	**Elman 1990 (RNNs)** ***also see:*** **Hochreiter 1997 (LSTM)**
Thursday, October 8, 2026				What networks get wrong: adversarial examples	**presentation**	Szegedy 2014 (Adversarial examples)
Tuesday, October 13, 2026				Transformers	**presentation**	Vaswani 2017 (Transformers)
Thursday, October 15, 2026				Multimodal representation learning: CLIP	**presentation**	Radford 2021 (CLIP) *also see:* Chen 2020 (SimCLR)
Tuesday, October 20, 2026		**HW4**	**HW3 + Quiz 3**	**Image generation**	**david**	

Thursday, October 22, 2026				Midterm review	david	
Tuesday, October 27, 2026			**Midterm**	**Midterm exam**	**midterm**	
Thursday, October 29, 2026			Project titles and teams due	Adversarial generation: GANs	**presentation**	Goodfellow 2014 (GAN)
Tuesday, November 3, 2026				Variational Autoencoders: VAEs	**presentation**	Kingma 2013 (VAE)
Thursday, November 5, 2026				Normalizing Flows	**presentation**	Dinh 2017 (Real NVP)
Tuesday, November 10, 2026				Diffusion Models	**presentation**	Sohl-Dickstein 2015 (Diffusion) *also see:* Ho 2020 (DDPM)
Thursday, November 12, 2026				Large language models	**presentation**	Brown 2020 (GPT-3)
Tuesday, November 17, 2026				Preference learning: RLHF and DPO	**presentation**	Rafailov 2023 (DPO) *also see:* Ouyang 2022 (RLHF)
Thursday, November 19, 2026			**HW4 + Quiz 4, project abstracts due**	**How to do research**	**david**	
Tuesday, November 24, 2026		HW5		RL with verifiable rewards: GRPO	david	Shao 2024 (GRPO)
Thursday, November 26, 2026	*Thanksgiving, fall break*					
Tuesday, December 1, 2026				Reasoning models: RL with verifiable rewards	**presentation**	DeepSeek 2025 (R1, RLVR)

Thursday, December 3, 2026			Project reviews due	Mechanistic interpretability	david	Meng 2022 (ROME)
Tuesday, December 8, 2026			**HW5 + Quiz 5**	**TBD**		
Thursday, December 10, 2026			Poster day		final projects	
Monday, December 14, 2026			final project report due (midnight)		final projects	

Student-Led Paper Discussions

Class participation and paper presentations: 50 points. In-class paper presentations and discussion.

Semester Overview: The centerpiece of the course is a series of student-led discussions. Throughout the semester, you will read and we will discuss 19 seminal research papers. Thirteen of those lectures will begin with a student-led discussion of one of the selected papers. Each of those student-led discussions is led by a panel of **four** students, who take four different roles, with follow-up questions from the rest of the class (*the questioners*). (With four to a panel everybody has a distinct job and time to speak.) **You will present on exactly one panel during the semester**, so the rest of your participation grade comes from submitting nightly reading questions and from discussion that your questions spark. These papers and their respective dates are pre-scheduled & can be seen in timeline schedule of the course structure.

Points Breakdown & To Dos (50 Points):

1. **Nightly Questions**: The night before each class that has an assigned paper, read the paper and submit a question on Canvas, with a sentence on why you think it matters. There are **19** such classes, 13 led by student panels. Prof. Bau will curate questions that students submit to drive the discussion by the panel, so when you submit a nightly question it is a way to put a question to both the class and Prof. Bau. Each question earns 1 point, for a total of **19 Points**. In several of the classes, a second paper is labeled *also see*, which you are also encouraged to skim. Submit reading questions about whatever you actually want to know. The main reading is the obvious place to start, but a related paper, the connection to other research or industry work, something from an earlier week, or something from outside the syllabus entirely are all fine. It is one question per class day, not one per paper. The nightly questions are part of your participation grade, so do not skip them.
2. **In-Class Interactions:** Prof. Bau will call students during any lecture, including asking you to present your nightly question, or to respond to a discussion in class; all together, class participation is worth **10 points.**

3. **Paper Presentation & Review**: Once during the semester you will be assigned one of the four roles below, to review and present one aspect of a paper, which will let you read that paper from a particular angle. Your paper presentation is a collaboration with three other students on your panel; your team will need to create an interactive visualization and a slide deck to spark the discussion (more on that below). Leading this discussion well earns up to **21 Points**.

Each student works with a group to help lead the discussion of **one** paper during the term, playing one of four roles on a team of four.

The four roles (one student each), in the order they present:

1. **Archaeologist**: Give the historical context and the biography of the authors. Explain the ideas that came before this paper, and what problem it was answering, and what intellectual impact it had after it was published. Explain the biography of the authors: who wrote it, where they were and who they were working with at the time, what they had done before, and what they went on to do.
2. **Diagrammer**: Create a slide that visually explains the method of the paper. And use it to explain the technical meat of the paper. What do the key formulas mean? How does the architecture work? What should you take away from the paper's findings?
3. **Demo Presenter**: Present a working technical artifact that *teaches* the paper's central idea, and run it live for the class. The whole panel builds the demonstration together; the demo presenter is the one who presents it, which means you have to understand it well enough to be asked about it. Details below.
4. **Reviewer**: Address the standard NeurIPS review questions about the paper. Write a real peer review (put the review document in your paper's folder in the shared drive), with suggestions to the authors on what seemed weak or strong to you about the paper. Put a summary of your review on your slide. You go last, and your review opens the discussion.

Panels that are not exactly four. Depending on enrollment, a few panels will be uneven. A panel of **five** puts a second student on the demonstration. A panel of **three** merges the Archaeologist and the Reviewer into one role, the ***Academician***, who opens the session with the context and closes it with the review.

Format & Preparation:

- **Duration:** Each paper discussion will last 45 minutes

- **Roles & Sign-up:** Fill in the sign-up form before **September 15, 2026**. Rank the papers you would most like to present and the roles you would prefer. We will try to honor your preferences, but with one panel per student some adjustment is inevitable.

- **Demonstration (built by the whole panel, presented by one):** Every panel must bring a short **interactive demonstration** of the paper's central idea — ideally something with a slider or a text box, so the class can change an input and watch what happens. A Colab notebook is the usual form. Building it is the panel's shared job: all four of you should have run it and told the *Demo Presenter* what was confusing. The Demo Presenter runs it in front of the room and answers for it. The aim is not a reproduction of the paper's results. It is one accurate, live illustration of the mechanism, small enough to run in the room, that leaves the class understanding something they did not understand before.

Use an LLM coding assistant to build it, and say so.

You are graded on *what you chose to show* — which mechanism from the paper you picked, what a viewer can change, and what they see when they change it — and on whether the demonstration is *accurate and informative*. Accurate: it really does implement what the paper describes, so check it against the paper before you show it. Informative: someone who watches it comes away understanding the mechanism better than they did. A demo that gets either of those wrong teaches the class the wrong thing.

On the demonstration slide, say briefly what the panel asked the assistant for and what you had to fix.

Some papers are about phenomena that only appear at a scale you cannot run in class. For those, demonstrate a *scaled-down analogue*, or make the paper's own figures interactive — let the class move along a curve the authors published. If you are not sure what would work for your paper, ask early.

- **Slide Deck & Materials Submission**: The panel submits its materials **one night before** the scheduled discussion. This includes:

- The Google Slides deck, with a slide from each of the four panelists. Everything goes in your paper's numbered folder in the shared presentations drive.

- The panel's interactive demonstration, as a link that runs from a fresh session on somebody else's computer. We will post it for the class, so anyone can open it and play with it afterwards. A Colab notebook is the usual form; a hosted page is fine too.

- The Reviewer's detailed review, uploaded to the same folder

- A question about the paper for the broader class (*submitted as your nightly question on Canvas*), and an acknowledgment of which role you are playing

Suggested demonstrations, by paper. These are ideas we think could be insightful, not requirements — if you have a better idea for your paper, propose it. Each one gives the class something to change and something to watch happen.

PAPER	AN INTERACTIVE VISUALIZATION THAT COULD BE INSIGHTFUL
Glorot 2010 (initialization)	A 30-layer MLP with a slider for the initial weight scale and a dropdown for tanh, sigmoid or ReLU. Show the activation histogram at every layer as a strip across depth, plus the gradient norm per layer. The insight: slide the scale and watch activations collapse toward zero or saturate, and see that the Xavier and Kaiming values are exactly where the variance stays flat with depth.

LeCun 1989 (LeNet)	A canvas where the class draws a digit, feeding a small LeNet, with every convolutional feature map shown live as the stroke is drawn. Add a slider that shifts, scales and rotates the drawn digit while the prediction and the maps update. The insight: what a convolution and a pooling layer actually compute on a real input, and that tolerance to a shift is built into the architecture while tolerance to a rotation is not.
He 2016 (ResNet)	Two networks side by side, plain and residual, with a depth slider from 2 to 60 layers. Plot the gradient norm arriving at each layer, and a short training curve for each. The insight: the plain network's gradient decays geometrically with depth while the residual one stays flat, and the plain network's *training* error gets worse as it gets deeper, which is the paper's actual headline.
Szegedy 2014 (Adversarial examples)	A pretrained image classifier with one photograph and a slider for the perturbation size. Show the original, the amplified perturbation, and the perturbed image side by side, with the top few class probabilities under each, and a button to target a class of the viewer's choosing. The insight: the two pictures are indistinguishable to the room while the label has changed completely, and how small the perturbation can be and still work. Then show the same perturbation moved to a different image, or a different network, to see how often it still transfers.
Vaswani 2017 (Transformers)	Type a sentence and see the attention matrix of a small pretrained model, per head and layer; click a token to see what it attends to. A switch to zero out the positional embeddings. The insight: different heads doing visibly different jobs, and the structure collapsing when position information is removed.
Radford 2021 (CLIP)	A text box for candidate labels and a handful of images, with the image-text similarity matrix computed live, so the class can invent new labels and watch zero-shot classification change. Then break it: try a typographic attack, writing a word onto an image. The insight: one shared embedding space for both modalities, and how shallow its grip on the image can be.

Goodfellow 2014 (GAN)	A 2-D target such as two moons. Show the generator's samples over the discriminator's decision surface as a background heatmap, updating as it trains, with sliders for the D-to-G update ratio and the two learning rates. The insight: the boundary chasing the samples, and mode collapse induced on purpose by over-training the discriminator.
Kingma 2013 (VAE)	A VAE with a 2-D latent trained on MNIST. A draggable pad over the latent plane that decodes to an image live, plus a beta slider. The insight: the latent space as a map you can walk around, and what raising beta does to both reconstruction sharpness and how fully the prior's mass decodes to something plausible.
Dinh 2017 (Real NVP)	A 2-D density warped by coupling layers, with a slider for how many layers have been applied and a toggle for the mask pattern, overlaying the running log-determinant. The insight: an invertible warp building up one layer at a time, and what goes wrong when you stop alternating the mask, so one coordinate is never transformed.
Sohl-Dickstein 2015 (Diffusion)	A single t slider from 0 to T: forward noising on the left, the reverse trajectory from a trained model on the right, and the alpha-bar schedule plotted underneath with the current t marked. The insight: the two processes as mirror images, and the schedule as the object that ties them together.
Brown 2020 (GPT-3)	Few-shot prompting with a small local model. A box where you add in-context examples one at a time, with accuracy on a small task plotted live against the number of examples. Alongside it, make the paper's own scaling curves interactive: slide model size and watch the zero-shot and few-shot gap. This is one of the two papers whose real phenomenon needs more compute than a classroom has, so the right move is a scaled-down analogue plus the authors' published data.

Rafailov 2023 (DPO)	A small preference sandbox. Start with a handful of candidate responses and pairwise preferences over them, and fit a Bradley-Terry reward model live. Then show the same preferences trained directly into a policy by the DPO loss, and plot the DPO *implicit* reward ($\beta \log \pi/\pi_{\text{ref}}$) against the explicitly fitted one. The insight: the two rankings agreeing even though no reward model was ever trained in the second case, and then the β slider pulling the policy away from the reference until it starts to hack the preferences. Like GPT-3, this needs a toy scale to stay accurate.
DeepSeek 2025 (R1, RLVR)	Build on HW5.1's own tiny GRPO loop: sliders for the KL coefficient, the length penalty and the group size, plotting reward, the fraction of scratchpad-style answers, and a live sample of the model's outputs. The insight: the reward-hacking result from the homework, reproduced in front of the room.

Class Procedure (during presentation):

1. **Archaeologist** (2-3 mins): where the paper came from and who wrote it.
2. **Diagrammer** (2-3 mins): the method itself, on one picture.
3. **Demo Presenter** (5-10 mins): run the demonstration live, and take a suggestion from the class for what to change.
4. **Reviewer** (2-3 mins): what is and is not established by the paper.
5. **Open discussion** (the rest): Prof. Bau or TAs will select students in sequence to pose their previously submitted questions to the panel.

Professor Bau will keep time and guide the class through the Google Slides.

Homework and Quizzes

Homework hand-in: 5 points. Programming and calculation exercises.

There are five homeworks, jupyter notebooks to be worked through by students individually. In this age of LLM assistance, we will not be scoring your homework code in detail. Instead, **each completed homework you hand in on Canvas earns 1 point**, for a total of 5.

There is no deadline on this and no late penalty. Hand a homework in whenever it is done, at any point in the semester — including *before* it is assigned, if you want to work ahead. The dates in the schedule above are when

each homework is *meant* to be done, because that is what the quizzes are timed against, but the point does not depend on them. There are no late days to keep track of. Suggestion: hand in all the homeworks early.

Quiz 0: 1 point. On the first day of class you will sit a short, ungraded diagnostic covering the prerequisite material: linear algebra, multivariable derivatives, probability, and the machine learning you have already seen. **Correctness is not scored.** We collect it at the end of class, so there is nothing for you to hand in; the point is simply for being there and attempting every part. If you do not know something, the most useful thing you can write is "don't know".

Its only purpose is to tell me what this particular class already knows, so that I teach the right things. The catalog lists four different courses that satisfy this course's prerequisite and they cover different material, so nobody is expected to have seen all of it.

Homework quizzes: 50 points. Five short quizzes, 10 points each, taken in class.

There is one homework quiz per homework assignment, and **each quiz is given in class on the day that homework is due** (the dates are in the schedule above). Those days are deliberately days without a student paper presentation, so that the class is not carrying two things at once. Each quiz tests your knowledge of the methods and concepts in that homework. The quizzes are designed to be much easier for you if you have actually worked through and understood the homework yourself, so the best preparation for a quiz is to do the homework yourself. This is where the real incentive to keep up with the homework lies: handing the homework in is worth 1 point, but its quiz is worth 10.

Quiz 3 is on October 20, a week before the midterm. HW3 covers recurrent networks, gating and attention, which are on the midterm, so doing it is part of preparing for the exam. The midterm review is on October 22 and the exam is on October 27; bring your questions to the review.

HW1 is due and Quiz 1 is given on September 22, which is also the add/drop deadline. HW1 goes out on the first day of class, so by the deadline you will have worked through a homework and sat a quiz on it. That is the clearest sense of the course you can get while you can still act on it.

On each of those five days we will open the class with about ten minutes of informal discussion of the homework, before the quiz. Bring your questions. If that discussion happens to give away the answer to something on the quiz, so be it — that is a reward for being in the room and paying attention.

Quizzes must be taken in person, in class on their scheduled day, to earn points. There is no remote or online version.

Each quiz is worth 10 points, and whether a quiz counts is decided as follows:

1. A quiz that would pull your grade down is **dropped** — out of the course total as well as out of your points — so a quiz that goes badly does not count against you.
2. A quiz you **skip** counts as a zero, and that zero stays in the total.
3. Coming to a **make-up** drops the missed quiz instead, the same way a low quiz you sat would be dropped.

The reason for scoring quizzes this way is that we would much rather you came to class than stayed away worried about a quiz. Come and take it.

For example: if you sit all five quizzes and they all help you, your grade is out of the full 206. If two of them would drag you down, those two are dropped and your grade is out of 186. If you sat none of them but came to all five make-up discussions, all fifty quiz points would drop out and your grade would be computed out of 156.

Make-ups run for 14 days after the quiz. A make-up begins with a second sitting of the quiz, and is then graded together with a short discussion of that homework with a TA, rather than marked cold — so if there is something you did not follow, that conversation is the place to sort it out. Come to office hours or arrange a time with a TA. After 14 days the make-up closes and the zero stands. If you know in advance that you will miss a quiz, tell us before the date rather than after.

The catch is that **the quizzes must be done in-person in class**. Each homework quiz is given in class on its scheduled day whether or not you have handed in the homework yet. The quizzes are worth 50 points and the homework hand-ins are worth 5, so the main reason to do a homework on time is not to hand it in: the homework is the best way to learn the material for the homework quiz. If you have done the homework yourself and understand the concepts, the quiz should be much easier.

Midterm Exam

Midterm exam: 50 points. A written, closed-book exam about foundational methods, held in class. The starred items in the concept list below are the ones that may appear on it.

Final Project Presentations

Final Project: 50 points. An illustrated blog or webpage report plus an in-class presentation, about a research topic centered on a paper you choose. Done in teams of 2 or 3 students.

This is the paper presentation again, except that you choose the paper and your team plays all four roles. Your project must give the **context** for the work, a **technical explanation** of the method and what its key formulas and architecture mean, an **assessment** of what the paper does and does not establish, and an **interactive visualization** of the central idea. You may also include a research extension of the concepts in the paper you choose.

The interactive visualization is **required, not optional**. The same standard applies as for the paper demonstrations: it should be accurate and informative, and a reader should be able to change something and see what happens. Hand it in as a **link that runs from a fresh session on somebody else's computer**, and we will post it for the class so everyone can open it and play with it at their own machine. A Colab notebook is the usual form; a hosted page is fine too. The measure of the whole project is whether a reader finishes it understanding the topic better than they did.

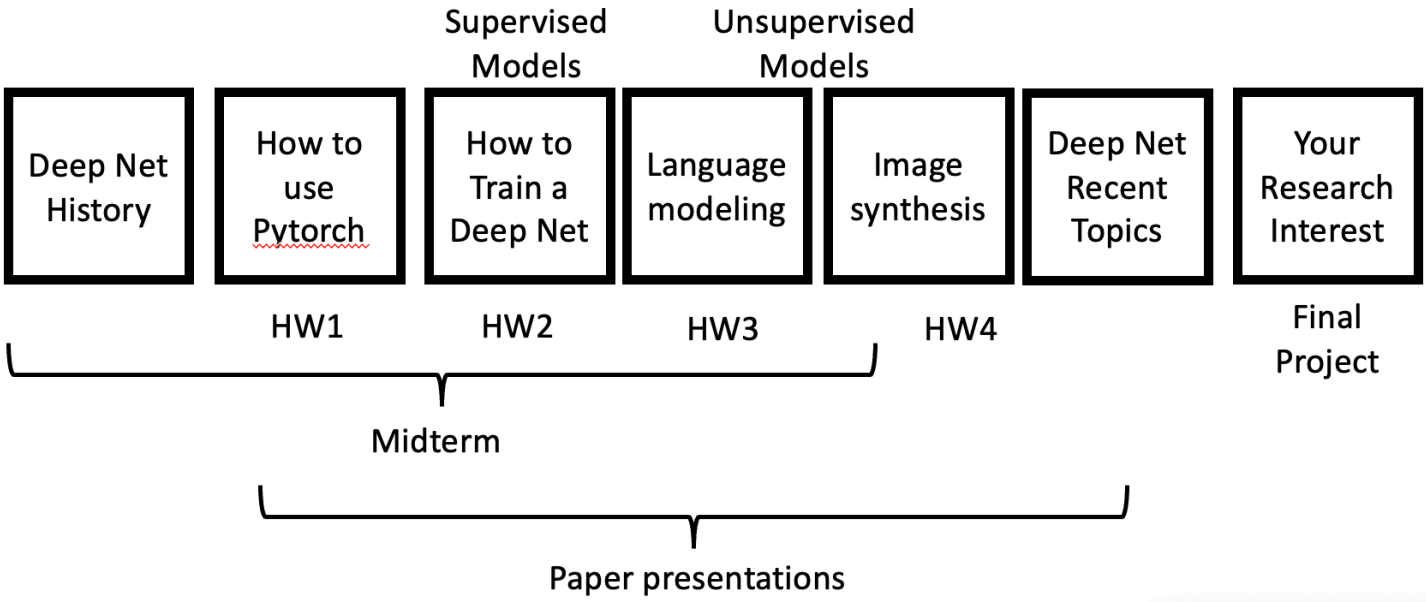
Collaboration Policy

Collaboration is allowed, but you should think about the problems yourself before discussing them with others or having an LLM solve it for you. When you do seek out help, we strongly advise you to find a fellow classmate to talk with and work together rather than just copying an answer. You will all learn much more by thinking collaboratively and explaining ideas to one another.

Academic integrity policy

Read through university's academic integrity policy and plagiarism.

Topic Map for the Course



List of important concepts to know after this course

*Starred items may be on the midterm exam.

The historical and intellectual evolution of deep network methods:

- From Ramón y Cajal to McCulloch-Pitts Neurons to Cybenko's Universal Approximation*
- Comparing Rosenblatt's Perceptron Learning to Rumelhart-Hinton's Backpropagation*
- Lettvin's Grandmother Neuron idea vs Hinton's Parallel Distributed Processing Model*
- Hinton-Salakhutdinov's Layerwise Pretraining vs Glorot-Bengio's End-to-End Training*
- From Hubel and Wiesel to LeCun's LeNet to Krizhevsky's AlexNet*
- From Elman's Structure in Time to GPT-3 Few-Shot Learning

How to use deep network tools (pytorch)

- GPU tensors*
- Autograd / backpropagation*
- SGD and Adam and other optimizers*
- Network modules*
- Fast data loading*
- Using pretrained models*

How to train a model deeper than a few layers

- The backpropagation algorithm*
- Vanishing and exploding gradient*
- Xavier or Kaiming initialization*
- ReLU vs Sigmoid vs Threshold nonlinearities*
- Residual connections*
- Batch normalization*

How to improve generalization

- Holdout tests*
- L2 Weight decay*, L1 regularization*
- Dropout*
- Early stopping*
- Data augmentation*
- Denoising*
- Weight sharing, invariants and equivariances*
- Double-descent

Learning to Learn

- Metalearning
- Prompting

- Reinforcement learning from human feedback (RLHF), Bradley-Terry preference models, direct preference optimization (DPO)
- Reinforcement learning from verifiable rewards (RLVR), GRPO

Doing research

- Reading papers
- Writing papers

*** Starred concepts may be on the midterm.**

How to design effective objectives

- Softmax and classification*
- Cross-Entropy loss, KL divergence, and distribution-matching*
- Autoencoders* and Denoising Autoencoders*
- L2 reconstruction loss and L1 median-finding loss*
- Contrastive learning*
- Variational Autoencoders
- Diffusion Models
- Normalizing Flows

How to input images

- Convolutional Layers*
- Pooling*
- AlexNet, VGG, and ResNet*

How to output images

- U-Nets*
- Perceptual losses
- Generative Adversarial Networks

How to understand and visualize a convolutional network

- Adversarial examples
- Gradient visualizations
- Saliency maps
- Feature visualizations and dissection

How to input and output text

- Word embedding*
- Recurrent networks*

- Language models*
- LSTM and GRU*
- Attention and Transformers*
- BERT, GPT, and T5

How to understand and visualize a language model

- Probability visualization
- Attention visualization
- Probing classifiers
- Logit lens
- Causal traces

What makes a good representation

- Semantic vector composition
- Disentanglement
- Transfer learning